You can create an outstanding, high-scoring Digital Humanities project using **zero code**. In DH, critical evaluation, research design, and humanistic analysis matter far more than writing code from scratch. Utilizing point-and-click browser auditing extensions, spreadsheet tracking, and GUI-based GIS tools is standard practice in humanities research.

The No-Code DH Toolkit

Category	Tool	How You Use It (Point-and-Click)
Web Auditing	**WAVE Extension** (Chrome/Firefox)	Instantly highlights WCAG errors, missing alt text, and structural bugs directly on the live web page.
Automated Auditing	**Chrome DevTools (Lighthouse)**	Built into Google Chrome (F12 → Lighthouse tab). Generates a 0–100 Accessibility score report in one click.
Accessibility Verification	**NVDA** (Windows) / **VoiceOver** (Mac)	Free built-in screen readers. Turn it on and use Tab and Enter keys to navigate the portal without a mouse.
Data Management	**Google Sheets / Excel**	Log error counts, page URLs, WCAG failure types, and score comparisons between GHMC and BBMP.
Spatial Mapping	**Datawrapper** or **QGIS**	Import ward boundaries or point locations to visually map web accessibility scores against urban zones or ward offices.

Step-by-Step No-Code Workflow for Topic 1

Step 1: Select Your Page Sample (10 Pages Total)

Pick 5 core public service portals each from **GHMC (Hyderabad)** and **BBMP (Bengaluru)**:

1. Home / Main Portal
2. Property Tax Payment Page
3. Citizen Grievance / Redressal Portal
4. Building Plan / Trade License Portal
5. Birth/Death Certificate Application Page

Step 2: Run One-Click Audits & Log Data

1. Open each URL in Google Chrome.
2. Open **Lighthouse** (F12 or Right-Click → *Inspect* → *Lighthouse* tab → Click *Analyze page load*). Record the overall **Accessibility Score (0–100)**.
3. Click the **WAVE Extension** button on your browser toolbar.
4. Record key numbers into Google Sheets:
    - Total Errors (Red icons)
    - Contrast Errors (Yellow/Black icons)
    - Missing Alternative Text (alt attribute) counts
    - Missing Form Labels

Step 3: Perform a 5-Minute Screen Reader Walkthrough

1. Turn on NVDA or VoiceOver. Put your mouse aside.
2. Attempt to perform one basic action using only Tab, Shift + Tab, and Enter (e.g., reaching the "Search" box or "Property Tax" input field).
3. Take screenshots wherever the cursor gets stuck, keyboard focus rings disappear, or the screen reader reads "unlabeled button."

Step 4: Map Spatial and Civic Context (No-Code GIS)

1. Use **Datawrapper** (a completely free, web-based, no-code mapping tool) or **QGIS** (desktop software with a visual interface).

2. Create a comparative spatial map:

    - Plot municipal head offices or ward boundaries across Hyderabad and Bengaluru.

    - Add interactive pins for each civic service, color-coded by their Lighthouse accessibility rating (e.g., Red = Score under 50, Yellow = 50–79, Green = 80+).

Step 5: Build Your PPT & Report Focus
Structure your project around the **humanistic implications** rather than technical implementation:

- **The Argument:** Show how poor digital accessibility acts as an invisible barrier that deprives visually impaired citizens of rights guaranteed by the *Rights of Persons with Disabilities (RPwD) Act, 2016*.

- **Visual Evidence:** Side-by-side bar charts comparing GHMC vs. BBMP scores, annotated screenshots showing screen reader failures, and your comparative GIS map.

Sighted vs. Visually Impaired Experience

Page Element	**Sighted Citizen Experience**	**Visually Impaired Citizen Experience (Screen Reader/Low Vision)**	**Proof of Exclusion**
Grievance Submit Button	Clicks the green icon.	Screen reader reads: *"Unlabelled Graphic"*.	**Absolute Denial:** User cannot submit a civic complaint.
Property Tax Due Date	Reads light-gray text easily.	Contrast ratio is 1.8:1; low-vision user sees blank space.	**Financial Penalty:** User misses payment deadline due to invisible text.

Form Error Validation	Sees the box outline turn red.	No ARIA-alert tag; screen reader announces nothing.	**Confusion:** User keeps clicking "Submit" with zero audio feedback.
Security CAPTCHA	Types distorted text.	No audio alternative or alt text provided.	**Total Lockout:** User cannot access the service independently.

How to Frame This as "Proof of Exclusion" in Your Report

In Digital Humanities, you prove exclusion by contrasting **intended public rights** with **technical execution failure**:

1. **Functional Exclusion vs. Cosmetic Flaws:** A slow website load time is a minor annoyance for everyone. A missing form label or image-only CAPTCHA is a **complete functional exclusion** for a blind citizen.

2. **Loss of Privacy & Autonomy:** When a civic portal forces a visually impaired citizen to hand their phone or bank details to a neighbor to read an unlabeled screen or CAPTCHA, the technology strips them of their right to private civic participation.

3. **Legal Non-Compliance:** India's *Rights of Persons with Disabilities (RPwD) Act, 2016* mandates that all government digital services conform to WCAG guidelines. Documenting these specific errors proves that civic bodies (GHMC/BBMP) are failing mandatory legal standards.

HOW TO ADAPT THIS PROJECT SPECIFICALLY FOR VISUALLY IMPAIRED PEOPLE:

Phase 1: Running Lighthouse Audits (Chrome DevTools)

Lighthouse evaluates a webpage against automated accessibility standards and generates an overall score from 0 to 100.

1. **Open an Incognito Window:** Press `Ctrl + Shift + N` (Windows) or `Cmd + Shift + N` (Mac) in Google Chrome. *Incognito prevents browser extensions from interfering with audit scores.*

2. **Navigate to Target Portal:** Load the public service page (e.g., GHMC Property Tax Payment or BBMP Citizen Grievance Portal).

3. **Open Developer Tools:** Press `F12` (or Right-Click anywhere on the page → select **Inspect**).

4. **Locate the Lighthouse Tab:** Click on the **Lighthouse** tab in the top menu of the DevTools panel. If hidden, click the `>>` icon to reveal extra tabs.

5. **Configure Settings:**

    - **Mode:** Navigation (Default)
    - **Device:** Desktop
    - **Categories:** Uncheck everything except **Accessibility**.

6. **Run Audit:** Click **Analyze page load**. Wait 15–30 seconds for Chrome to complete the audit.

7. **Extract Accessibility Metrics:**

    - **Overall Score:** Note the 0–100 circular score at the top.
    - **Filter Findings for Visual Impairment:** Scroll down to the *Passed Audits* and *Failing Audits* sections. Look specifically for:

        - `[aria-*] attributes do not match their roles` (Breaks screen readers).
        - `Background and foreground colors do not have a sufficient contrast ratio` (Blocks low-vision users).
        - `Image elements do not have [alt] attributes` (Blocks blind users).

8. **Save Report:** Click the three vertical dots (⋮) in the top-right of the Lighthouse panel → select **Save as HTML** or **Save as JSON** to keep proof for your report.

Phase 2: Running WAVE Audits (Browser Extension)

WAVE overlays visual icons directly onto the live web page, pinpointing exact element failures in the DOM.

1. **Install WAVE:** Add the **WAVE Evaluation Tool** extension from the Chrome Web Store or Firefox Add-ons.

2. **Trigger WAVE:** Navigate to your target GHMC or BBMP page and click the **WAVE icon** (`W`) in your browser toolbar.

3. **Analyze Summary Sidebar:**
    - **Errors (Red Icons):** Critical WCAG violations that completely block screen reader users.
    - **Contrast Errors (Black/Yellow Icons):** Text contrast failures that render content invisible to low-vision users.
4. **Isolate Specific Visual Exclusion Metrics (Details Tab):**
    - Click the **Details** tab in the WAVE sidebar.
    - Count occurrences of **Missing alternative text** (image without description).
    - Count occurrences of **Missing form label** (screen reader cannot identify what input box requires).
    - Count occurrences of **Empty button** (screen reader reads "Button" with no context).
5. **Evaluate Heading Hierarchy (Structure Tab):**
    - Click the **Structure** tab to check if the page uses proper `<h1>`, `<h2>`, `<h3>` tags. Visually impaired users use heading shortcuts to skim pages; if no headings are detected, document this as structural exclusion.
    -
6. **Capture Evidence for PPT:** Take screenshots showing the red WAVE error icons overlaying key functional areas of the page (like "Submit Payment" or "CAPTCHA" boxes).

Phase 3: Mapping Visual Exclusion (Datawrapper & QGIS)

Creating spatial maps demonstrates that digital barriers correlate directly with civic administration zones.

Option A: Datawrapper (Fast, Web-Based, No-Code)

1. **Prepare Spreadsheet Data:** Create a Google Sheet or Excel tab named `Datawrapper_Import` with these columns:

Zone / Office Name	City	Latitude	Longitude	Service Portal	Accessibility Score	Critical Barriers Found
Khairatabad Zonal Office	Hyderabad	17.4116	78.4624	Property Tax	42	Unlabeled Pay Button, No Audio CAPTCHA
Mahadevapura Zonal Office	Bengaluru	12.9904	77.6974	Trade License	38	Missing Alt Text, Low Contrast Text

2. **Create Map:** Go to datawrapper.de and click **Create new Map** → select **Symbol Map**.

3. **Import Data:** Select **Upload CSV** or copy/paste your spreadsheet columns directly into the text box.

4. **Customize Visual Markers:**

    - Set symbol color based on Accessibility Score: **Red** (Score < 50 / Severe Exclusion), **Yellow** (Score 50–74 / Moderate Exclusion), **Green** (Score 75+ / Accessible).

5. **Configure Tooltips:** Enable hover popups so that clicking a municipal zone marker displays the portal evaluated, exact WCAG errors, and screen-reader status.

6. **Export:** Click **Publish** to embed the interactive map link into your report or export a high-resolution PNG image for your PPT slides.

Option B: QGIS (Spatial GIS Layer Analysis)

1. **Download Civic Boundary Files:** Download open-source municipal ward/zone boundary data (GeoJSON or KML format) for GHMC and BBMP from public repositories like DataMeet or OpenCity.

2. **Import Spatial Layers:** Open QGIS → Click **Layer** → **Add Layer** → **Add Vector Layer** → Load your downloaded GHMC/BBMP `.geojson` or `.kml` file.

3. **Import Audit Data CSV:** Go to **Layer** → **Add Layer** → **Add Delimited Text Layer** → Load your accessibility audit CSV spreadsheet.

4. **Perform Attribute Join:**

    - Right-click the Boundary Vector Layer in the Layers panel → **Properties** → **Joins**.

    - Click the `+` icon. Join your Audit CSV to the Boundary file using a common field (e.g., `Zone_ID` or `Ward_Name`).

5. **Create Choropleth Map (Color-Coded Exclusion):**
6.

    - In Layer Properties, select **Symbology**.

    - Change dropdown from *Single Symbol* to **Graduated**.

    - Select `Accessibility_Score` as the value column.

    - Choose a red-to-green color ramp (Red = Low Score / High Digital Exclusion).

    - Click **Classify** → **Apply**.

7. **Export Final Map:** Click **Project** → **New Print Layout** to generate an annotated map export complete with a legend, scale bar, and title ("Digital Accessibility Barriers Across GHMC and BBMP Municipal Zones").